can be explained by that portfolio's market risk (beta), percentage in small stocks (size risk), and percentage in high book-to-market stocks (value risk). Over the long term, U.S. small-cap stocks have achieved a return premium over large-cap stocks, and value stocks have achieved a return premium over growth stocks.

Several independent studies have confirmed that the behavior of widely diversified international stock portfolios is consistent with the three factors found in U.S. stock portfolio returns.[1] Using DFA indexes, Table 7-5 quantifies the historic premium paid to international small-cap stocks over international large-cap stocks and the premium paid to international value stocks over international growth stocks. An analysis of MSCI index data produced similar results.

Between 1999 and 2009 the international markets exhibited both a size premium and a value premium. Based on the DFA indexes, the international size premium annualized at about 3.9 percent over the period and the value premium was 4.7 percent. There was over a 2.0 percent annualized premium for value stocks and small-cap stocks based on MSCI EAFE indexes going back to 1975.

TABLE 7-5

International Size and Value Premiums, 1999–2009

Size Premium	DFA International Large-Company Index	DFA International Small-Company Index
Annualized return	4.8%	8.7%
Standard deviation	17.5%	18.5%
Value Premium	**DFA International Large-Company Value**	**DFA International Small-Company Value**
Annualized return	8.0%	12.7%
Standard deviation	21.4%	18.1%

The international size and value premiums have been documented in many countries during independent time periods. This leads researchers to believe that the two premiums are not an anomaly. Researchers are becoming more convinced that size and value premiums represent